Master's Thesis

Online Imputation of Missing Data in Financial Time Series

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Abstract

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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. In every one of these settings a practitioner or a researcher faces the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

This question is central for market makers and risk managers. For instance, an emerging markets trader could be asked to make a market for an highy illiquid instrument that has not traded for several hours or days. Depending on the instrument, several approaches are possible: for a bond, she could interpolate the missing yield from the rest of the curve; for a derivative, she could use her own pricing model and hope that the imputs are not too stale.[1] A portfolio manager could be asked to compute the optimal weights of a portfolio that contains illiquid assets, and she must decide how to treat the missing returns in the covariance matrix.

The simplest solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve

[1] Note that here she is faced by the same problem, i.e. factors (as α, β, ρ in SABR) that may have not been observed in a some time; or equivalently missing points in the implied volatility surface

as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the algorithm.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how different it is from the case where all data is observed.

1.2 Why missingness in finance is hard

The modern theory of missing data originates with Rubin (1976), who classifies the mechanism that generates the gaps. Let the complete data be partitioned into observed and missing parts, $X = (X_{\text{obs}}, X_{\text{mis}})$, and let R denote the matrix of missingness indicators, governed by a conditional law $P(R \mid X, \phi)$. The mechanism is *missing completely at random* (MCAR) if R is independent of the data, *missing at random* (MAR) if it depends only on the observed part X_{obs}, and *missing not at random* (MNAR) if it depends on the unobserved values X_{mis} themselves. The practical importance of this taxonomy lies in the *ignorability* result: likelihood-based and Bayesian inference may ignore the missingness mechanism if and only if the data are MAR and the data and mechanism parameters are distinct Rubin (1976); **?**. This is the condition that licenses the expectation–maximisation algorithm Dempster et al. (1977) and multiple imputation Rubin (1987) without explicitly modelling R.

Financial data violates the convenient end of this taxonomy. Missingness in returns and fundamentals is typically MNAR and strongly structured: it clusters in the cross section and persists over time, and the very fact that a value is missing is informative about that value. Returns themselves depend on whether data are missing **?**. Two consequences follow directly and motivate the methods studied here. First, complete-case analysis conditions on survival and on disclosure, producing a sample that is not representative of the population of interest. Second, unconditional mean imputation, even where it leaves first moments unbiased, attenuates estimated variances and covariances toward zero. Because mean–variance portfolio selection and risk measurement consume the covariance matrix Σ—and, through the optimiser, its inverse Σ^{-1}—this attenuation corrupts exactly the object the downstream model depends upon.

1.3 From imputation to estimation: the downstream view

The clearest illustration of the downstream view, and the running example of this thesis, is mean–variance portfolio selection **?**, in which the optimal risky portfolio is proportional to $\Sigma^{-1}\mu$ for a mean vector μ and covariance Σ. It is well established that plugging sample estimates into this expression performs poorly out of sample: the optimiser behaves as an "error maximiser", loading on assets whose moments are most favourably mis-estimated **??**. The estimation-error problem is severe enough that a naive equally weighted portfolio is hard to beat **?**, and a large literature responds with regularisation: shrinkage of the covariance matrix **??** or equivalent weight constraints **?**.

Imputation enters this picture not as a separate preprocessing step but as a further source of regularisation. Filling a return panel with a low-rank or factor-based reconstruction shrinks the estimated covariance toward a factor structure—the same target used deliberately by shrinkage estimators. The choice of imputation method and the choice of covariance estimator are therefore not separable, and the error that imputation introduces propagates through Σ^{-1} into the portfolio weights and ultimately into realised performance. This is why reconstruction accuracy is an inadequate criterion: it ignores how the imputation interacts with the conditioning of the downstream estimator. The natural evaluation is instead a two-layer one, comparing methods both on reconstruction error and on a downstream objective such as realised portfolio variance, Sharpe ratio, or tail risk, with particular interest in the configurations where the rankings reverse.

Mean–variance optimisation is only the leading example. The same impute-then-estimate logic governs risk-model estimation and the computation of Value-at-Risk and Expected Shortfall; the estimation of factor models and cross-sectional return predictors from panels of firm characteristics **??**; volatility forecasting; macroeconomic nowcasting from mixed-frequency, ragged-edge data; the construction of yield curves and implied-volatility surfaces from sparsely observed grids; and credit-risk and ESG scoring, where the underlying data are both heavily and informatively missing. In each case the relevant question is identical: how does the imputation choice change the estimate that the practitioner actually uses.

1.4 Sparse and illiquid markets

The problem sharpens in emerging and illiquid markets, where it also changes in character. The dominant difficulty there is often not an explicit missing cell but *staleness*: an asset that does not trade still carries a recorded closing price—yesterday's—so the panel can appear fully observed while being silently contaminated by repeated prices and spurious zero returns. The missingness is in part *latent*, inferred rather than flagged, which must be detected before it can be addressed, for instance through the proportion of zero-return days and related liquidity proxies **??**. Whether a stale observation should be treated as observed or converted to missing is itself a modelling decision with downstream consequences.

Illiquidity also intensifies the statistical biases of interest. Non-trading is more likely in adverse states of the world, so the missing observations are disproportionately the extreme ones, which inflates estimated means and understates downside risk. Non-synchronous prices bias correlations and betas downward and understate volatility, so that a portfolio optimiser fed stale data systematically understates risk and overstates diversification; standard corrections estimate exposures using leads and lags **??**. Because the characteristic failure mode is the under-statement of risk, evaluation in this setting should emphasise realised downside measures rather than in-sample Sharpe ratios alone. Finally, illiquidity yields a testable prediction that the thesis examines: since the time series of an illiquid asset is largely uninformative, cross-sectional and low-rank methods, which borrow strength from contemporaneous liquid peers, should outperform time-series fillers by a wider margin in emerging markets than in developed ones.

1.5 Research questions

The thesis is organised around the following questions.

1. To what extent does the ranking of imputation methods by reconstruction accuracy differ from their ranking by downstream model quality, and under what conditions do the two diverge?
2. How do these rankings depend on the missingness mechanism (MCAR, MNAR, and clustered or persistent patterns) and on the fraction of data that is missing?
3. How should the impute-then-estimate pipeline be adapted to emerging and illiquid markets, where staleness and non-synchronous trading dominate, and does the predicted advantage of cross-sectional methods materialise there?

4. How can the uncertainty introduced by imputation be propagated honestly into the downstream estimate rather than ignored?

1.6 Contributions

In addressing these questions the thesis makes the following contributions.

1. It formalises and empirically documents the gap between imputation accuracy and downstream utility, exhibiting concrete rank reversals in which the most accurate imputation does not yield the best downstream estimate.

2. It develops a controlled experimental methodology in which a fully observed panel serves as ground truth (an *oracle*): entries are deleted under a chosen mechanism, imputed, and scored on both layers, with the missingness mechanism treated as an explicit experimental factor.

3. It enforces point-in-time discipline throughout, imputing and estimating using only information available at each date, and it quantifies the look-ahead bias incurred when this discipline is relaxed.

4. It propagates imputation uncertainty into the downstream estimate via multiple imputation and Rubin's combination rules Rubin (1987); **?**, in contrast to single imputation, which understates that uncertainty.

5. It adapts the framework to illiquid and emerging markets through staleness detection, mechanism-aware masking, and risk estimators robust to non-synchronous trading, and tests the resulting predictions empirically.

1.7 Outline of the thesis

The remainder of the thesis is organised as follows. Chapter 1 has motivated the problem and stated the research questions. The following chapter reviews the relevant literature, spanning the statistical theory of missing data, low-rank matrix completion **??**, and the econometrics of portfolio estimation error. The subsequent chapter sets out the methodology: the data, the missingness mechanisms, the imputation methods under comparison, the downstream estimators, and the two-layer evaluation. A further chapter reports the empirical results across mechanisms and missing rates, including the emerging-market case. The final chapter discusses the findings, their limitations, and directions for future work.

Chapter 2

Methodology

Chapter 3

Literature Review

Chapter 4

Imputation Methods

Chapter 5

Results and Discussion

Chapter 6

Conclusion

6.1 Summary

6.2 Limitations

6.3 Future Work

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